Bill of Materials for a Physical Build

Non-Invasive Bone Fracture Detection by UWB Microwave Imaging

Companion to the submission report. Every specification is traced to the simulation that fixed it.

The project is purely virtual by design, and nothing listed here has been bought. This is what a physical build would need, with each specification traced to the simulation that fixed it rather than to a generic shopping list. Cost figures are the proposal's indicative Indian-supplier numbers (Robu, Element14 India, Mouser India, Amazon.in) and **must be re-quoted before purchase**; several are a year old.

Configuration	Approximate total
With a department VNA (recommended)	**INR 23,800**
With the LiteVNA-64 fallback	**INR 32,800**

The one thing the proposal could only guess at — whether the cheap VNA fallbacks actually work — is now measured (Section 2). The spreadsheet form, *docs/bill-of-materials.csv*, carries all three VNA options, so its column sum (INR 42,700) is not a configuration total.

1 The measurement instrument

The VNA decides viability. The proposal's framing stands: *the single question is whether the department owns a VNA reaching 8 GHz or above.*

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
VNA, preferred	1-port S11; sweep 3–11 GHz (antenna is matched 2.48–12.00 GHz); at least 201 points; output 0 dBm; dynamic range 95 dB or better at about 1 kHz IF bandwidth	Department lab VNA, 8–20 GHz	1	0	Check first; decides everything below
VNA, fallback 1	50 kHz – 6.3 GHz	LiteVNA-64	1	9,000	**Works**; measured below
VNA, fallback 2	To 4.4 GHz	NanoVNA V2 Plus4	1	7,000	Works on band; dynamic range untested
Calibration kit	SMA short / open / load, calibrated **at the antenna connector**, not at the VNA port	SMA SOL kit (often bundled with the LiteVNA)	1	1,500–3,000	Everything downstream is referenced to the antenna feed

Where the numbers come from. The matrix was solved over 3–11 GHz at 201 points. The error model assumes a **95 dB** noise floor below the strongest return, which a lab VNA at 1 kHz IF bandwidth meets and a LiteVNA (roughly 70 dB at its top end) does not. **0 dBm** is the adopted transmit power from the exposure assessment: at +10 dBm the 6–11 GHz absorbed power density sat within 1.8× of the ICNIRP public limit; at 0 dBm every limit clears by at least 18×, and the detector has 40.8 dB of margin to spend (Section 7).

2 Do the cheap VNAs work? Measured, not guessed

The proposal labelled the LiteVNA “band-limited; degrades resolution” and the NanoVNA “minimum viable, marginal”. Those were guesses made before the matrix was solved. Every solved cell stores the full 3–11 GHz response, so each fallback can be scored on exactly the data it could have acquired, with no new simulation (*scripts/vna_fallback_study.py*, same pipeline and detector as Stage 7).

VNA option	Usable band	Bandwidth	Range res.	AUC	Class separation
Department VNA	3.00–11.0 GHz	8.00 GHz	3.67 mm	1.0000	**40.8 dB**
(first antenna, for scale)	4.35–11.0 GHz	6.65 GHz	4.42 mm	1.0000	17.9 dB
LiteVNA-64	3.00–6.30 GHz	3.30 GHz	8.91 mm	1.0000	**51.5 dB** at 95 dB dynamic range; **36.4 dB** at 70 dB
NanoVNA V2 Plus4	3.00–4.40 GHz	1.40 GHz	21.0 mm	1.0000	**43.0 dB**, band only, at 95 dB dynamic range

Verdict: both work, and on the band alone the LiteVNA beats the department VNA. Removing the upper band *raises* the separation, 51.5 dB against 40.8, because the frequencies above 6.3 GHz contribute little through 6 mm or more of overlay and mostly add image background the detector does not want; the low band carries the penetration (Section 6). By overlay, the LiteVNA band gives 64.9 / 66.6 / 65.2 / 51.5 dB at 0 / 3 / 6 / 10 mm and the NanoVNA band 54.8 / 57.3 / 60.5 / 45.9 dB. The classes do not overlap at any overlay for either.

Dynamic range, measured separately. The band study keeps the error model's 95 dB noise floor, which the cheap VNAs do not have. Re-scoring the LiteVNA band at 70 dB (*scripts/vna_dynrange_study.py*, 576 realisations) gives AUC 1.0000 and **36.4 dB** separation: 50.2 / 49.1 / 50.0 / 36.4 dB by overlay. The 25 dB worse floor costs 15 dB of margin, nearly all of it at the 10 mm overlay, and what remains is within 5 dB of the department VNA's full-band 40.8. A LiteVNA-64 is therefore a defensible instrument for this experiment, not a compromise. The same test has not been run for the NanoVNA band; its 43.0 dB is the band-only figure at 95 dB and should be read as an upper bound on what that instrument would deliver.

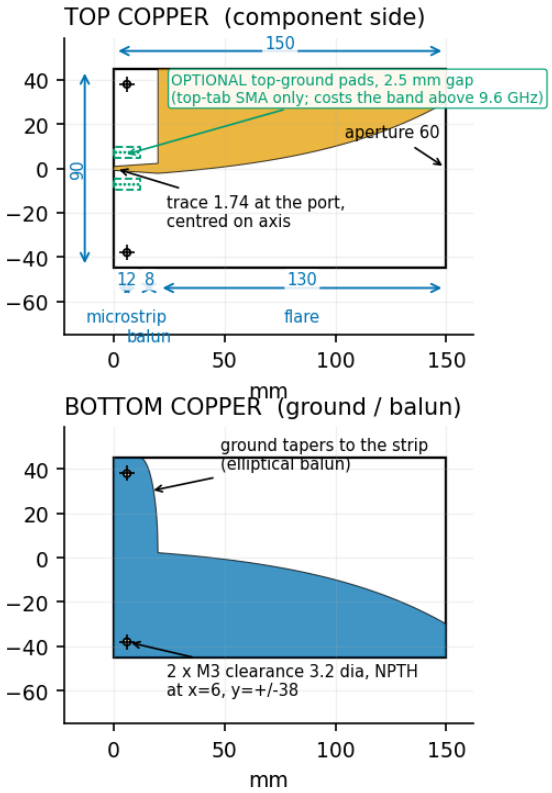
3 The antenna

Design files are in *figures/*: *vivaldi.dxf* (4 layers, mm, origin at the feed edge), *vivaldi_antenna.step*, *vivaldi_fab_drawing.pdf* (A4, both faces dimensioned) and *vivaldi_geom.json*. Rebuild with *python scripts/make_antenna_fab.py*.

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Substrate	Rogers RO4350B, 0.762 mm (30 mil), 1 oz Cu, 100 × 150 mm; the antenna is 150 × 90 mm, one board per antenna	RO4350B 0.762 mm	2	3,500	FR4 alternative INR 300; costs 2.4 dB at 9 GHz and 3.6 dB at 11 GHz (Stage 5)
Fabrication	Etch both faces from *vivaldi.dxf*; four 3.2 mm mounting holes at (6, ±38) mm	PCB house, or in-house etch	2	3,000	One spare
SMA connector	**End-launch, 18 GHz rated**; port trace is 1.74 mm (50 ohm)	L-com RFPCB-SMA-F S18G-55 (solder, cut for 0.031 in board) or Southwest 292-04A-9 (solderless clamp, 0.010 in pin)	4	1,600	**Do not use cheap 6 GHz parts.** No extra pads needed; see note
Mounting	M3 nylon standoffs and screws for the four holes	M3 × 10 nylon	8	200	Keeps metal out of the near field

The antenna as designed. Antipodal Vivaldi, growth a·T = 2.0, 60 mm aperture, **tapered feed 4.50 mm at the flare to 1.74 mm at the port**. The taper is what took the match from 4.35–12.00 GHz to **2.48–12.00 GHz**: 99.69% of 3–11 GHz below –10 dB, worst –9.98 dB at one point, median Re(Z_{in}) 45.6 ohm (Section 5). Gain is 7.02 dBi at 3 GHz, 11.9–12.5 dBi across 5–9 GHz and 9.3 dBi at 11 GHz; the 8–10 dBi target is met from about 4 GHz up and **not at 3 GHz**, which is aperture-limited and was shown not fixable by resizing within the board.

Connector note, from the fabrication drawing. Both named parts were checked against their datasheets for the 0.762 mm board. An end-launch lands its pin on the 1.74 mm trace at the board edge and grounds to the full-width plane on the other face, so **no pads are drawn and none are needed**. The Southwest part's 0.0635 in (1.61 mm) dielectric is within 8% of the trace width, which is why it launches cleanly. The proposal's INR 400 per connector assumes a generic 18 GHz end-launch; the named Southwest part is a premium item at roughly ten times that — price it before choosing. Top-ground-tab connectors are supported only with a 2.5 mm or wider pad gap and cost the band above 9.6 GHz; avoid them.



Fabrication drawing of the antipodal Vivaldi, dimensioned panels only; the full sheet with the connector and pad notes is figures/vivaldi_fab_drawing.pdf. Both faces, 150 × 90 mm on RO4350B 0.762 mm, feed tapered from 4.50 mm at the flare to 1.74 mm at the port.

4 Cabling

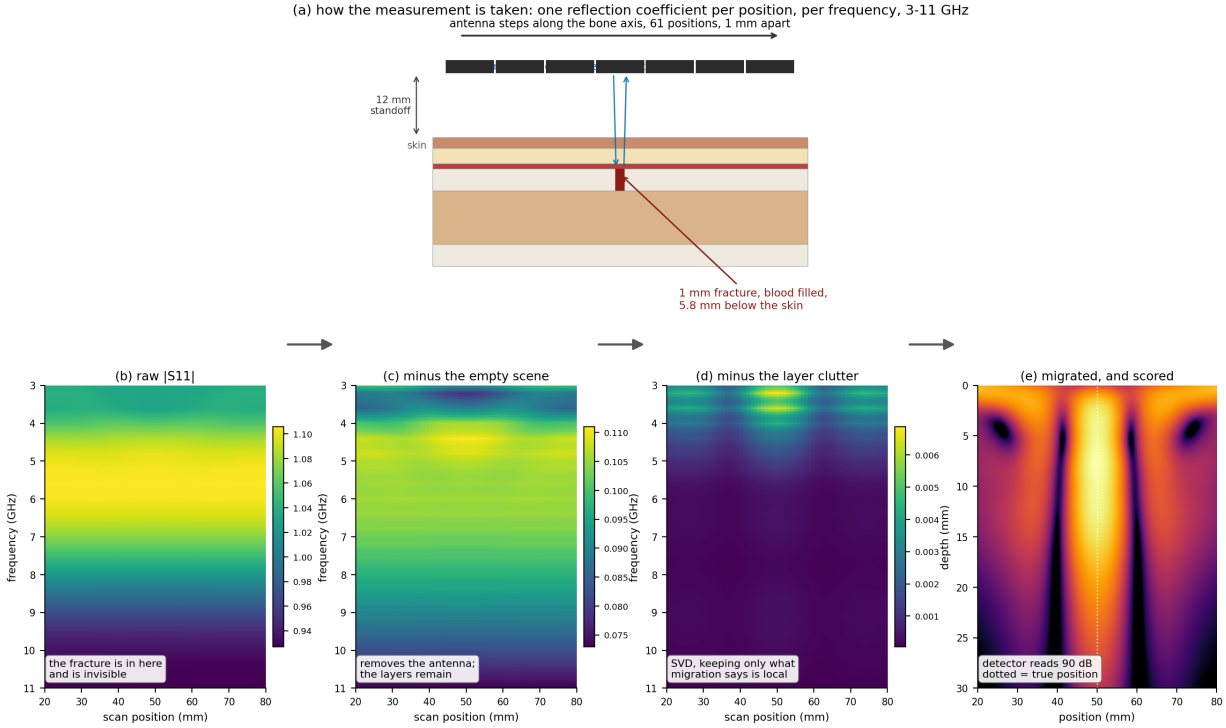
Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Test cable	Phase-stable, SMA–SMA, 18 GHz, 60 cm	Mini-Circuits or Pasternack phase-stable	1	4,500	**Secure it** — flexing changes phase
Port saver	SMA female–female adapter on the VNA port	SMA F–F, 18 GHz	2	500	The VNA connector outlives the project; the saver does not

The error model's cable term is 1 fs RMS delay jitter, i.e. a cable that does not move during a scan. Route it along the stage with slack at a single fixed bend, not dragged.

5 The scan stage

Migration sums phase coherently across positions, so position error is phase error: 0.5 mm at 10 GHz in air is about 60 degrees. The error model assumes **100 µm RMS** stage jitter and the result tolerated it; build to that.

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Rail	At least 60 mm of scan travel (the matrix used 61 positions at 1 mm, 20–80 mm); 200 mm gives clearance for the 150 mm antenna	2020 aluminium extrusion, 200 mm	1	2,800	Includes lead screw
Lead screw	8 mm, **anti-backlash nut**	T8 with anti-backlash brass nut	1	(incl.)	Backlash is a systematic position error migration cannot remove
Motor and driver	NEMA 17, A4988, 12 V 3 A supply	Standard 3D-printer parts	1	1,400	1 mm per step command; settle 200 ms before each sweep
Home switch	Mechanical limit switch at one end	Microswitch	1	(incl.)	Repeatable zero; required for empty-scene subtraction to line up
Controller	Serial command: step, settle, trigger	Arduino Nano or ESP32	1	600	The laptop drives the VNA over USB; the controller only moves the stage



The measurement the stage and fixture must reproduce: 61 antenna positions at 1 mm along the bone axis, 12 mm standoff, one S11 sweep per position, then empty-scene subtraction, clutter removal and migration.

6 Fixture

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Antenna holder	**12 mm standoff** from aperture to skin, held to **±0.3 mm**	Acrylic or 3D-printed bracket on the stage carriage	1	(in 2,500)	standoff_sigma = 300 µm in the error model
Limb / phantom rest	Holds the specimen with the bone axis parallel to the rail	Acrylic cradle	1	(in 2,500)	Tilt shows up as a sloping skin line
Phantom tank	For casting layered phantoms around the bone	Acrylic box with lid	1	(in 2,500)	Cast overlay slabs separately (next section)
Frame	Mounts rail, holder and rest in one rigid assembly	2020 extrusion and brackets	1	2,500 total	Rigidity is what makes the 100 µm stage spec achievable

7 The phantom and specimens

The simulated layer stack is the casting target. Thicknesses are from *bonesim/geometry/phantom.py*; dielectric targets per tissue are in Section 3 (Gabriel four-Cole-Cole).

Layer	Thickness	Note
Skin	2.0 mm	Low oil fraction
Fat	3.0 mm	High oil fraction
Muscle	1.0 mm	High water
Near cortical wall	4.0 mm	The bone specimen itself
Marrow	10.0 mm	The bone specimen itself
Soft-tissue overlay	**0, 3, 6, 10 mm**	The four experimental conditions; cast as separate slabs laid over the skin layer

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Phantom materials	Oil-in-gelatin recipe (proposal 4.3)	DI water 50–70%, porcine gelatin 250 bloom 10–15%, kerosene or safflower oil 15–35% (the tuning knob), surfactant 1–2%, n-propanol 0–5%, p-toluic acid under 0.5%, NaCl 0–1%	–	1,800	Oil fraction sets permittivity 10–50; **measure each batch** with an open-ended coaxial probe on the VNA
Bone specimens	Mid-shaft, about 4 mm cortical wall, about 10 mm medullary cavity	Goat or bovine tibia, fresh from a butcher	6–12	900	Defat mechanically; **do not boil**; store in saline gauze at 4 °C
Fracture fill	Blood-mimicking fluid; haematoma is the real contrast	Saline-based blood mimic	–	(in 1,800)	The simulation used Gabriel blood

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Validation target	**6 mm brass rod at 20 mm depth**	Brass rod, 6 mm diameter, 100 mm	1	200	**Stage 4 gate: image this before any bone.** If migration cannot find it, nothing downstream is meaningful
Kerf tools	Kerfs of 0.2–2.0 mm were simulated; the saw sets what is achievable	Jeweller's saw, diamond wafering blade	–	(in 1,200)	Measure the actual kerf; the measurement is the ground truth
Metrology	–	Digital calliper, feeler gauges	–	(in 1,200)	Record every kerf and every cortical thickness as covariates
Consumables	–	Cling film, gloves, saw blades	–	1,200	–

8 Compute and software

Nothing to buy. The laptop that ran the analysis runs the acquisition.

Item	Note
bonesim package	Processing chain, detector, statistics — unchanged between simulation and bench
VNA control	scikit-rf and pyvisa for a lab VNA; LiteVNA and NanoVNA have serial protocols with Python libraries
Stage control	Serial to the Arduino or ESP32

The processing chain was built so that it cannot tell the analytic model, the FDTD solver and a real VNA apart: it sees a matrix of complex S11 and nothing else. That is the point at which the simulation hands over to the bench.

9 What the build would test that the simulation could not

In order of how much they threaten the result (Section 8):

The error model. 95 dB dynamic range, 100 µm stage jitter, 1 fs cable jitter and ±0.3 mm standoff are assumptions. The 40.8 dB margin is the budget the real bench has to stay inside. This is what a brass rod and a VNA measure on day one.

Real bone. Eight simulated specimens vary only in what the variation model perturbs. Goat tibiae vary in everything.

Fracture position. Every simulated fracture is at x = 50 mm. A real cut is wherever the saw went, which tests localisation rather than separability.

Three dimensions. The bench is 3D for free. The 3D model (*bonesim/geometry/cylindrical.py*) is built and verified but unsolved.